

IT-Simplera Solutions



Week 3 Task Report

Enterprise Multi-Protocol Routing Challenge (OSPF Multi-Area, Static Routing & RIP v1/v2)

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Enterprise Multi-Protocol Routing Challenge (OSPF Multi-Area, Static Routing & RIP v1/v2)

1. Network Design & Core Concepts

1.1. Open Shortest Path First (OSPF) Multi-Area

- **Definition:** OSPF is a link-state routing protocol that uses the Shortest Path First (SPF) algorithm to find the best path to a destination. Multi-Area OSPF divides a large routing domain into smaller, logical groups (Areas) to localize routing updates and minimize SPF calculation overhead.
- **Explanation:** By restricting topology changes to a specific area (e.g., Area 1 or Area 2) and requiring all areas to connect through a central backbone (Area 0), the network saves memory and processing power on routers.

1.2. Routing Information Protocol (RIP v1/v2)

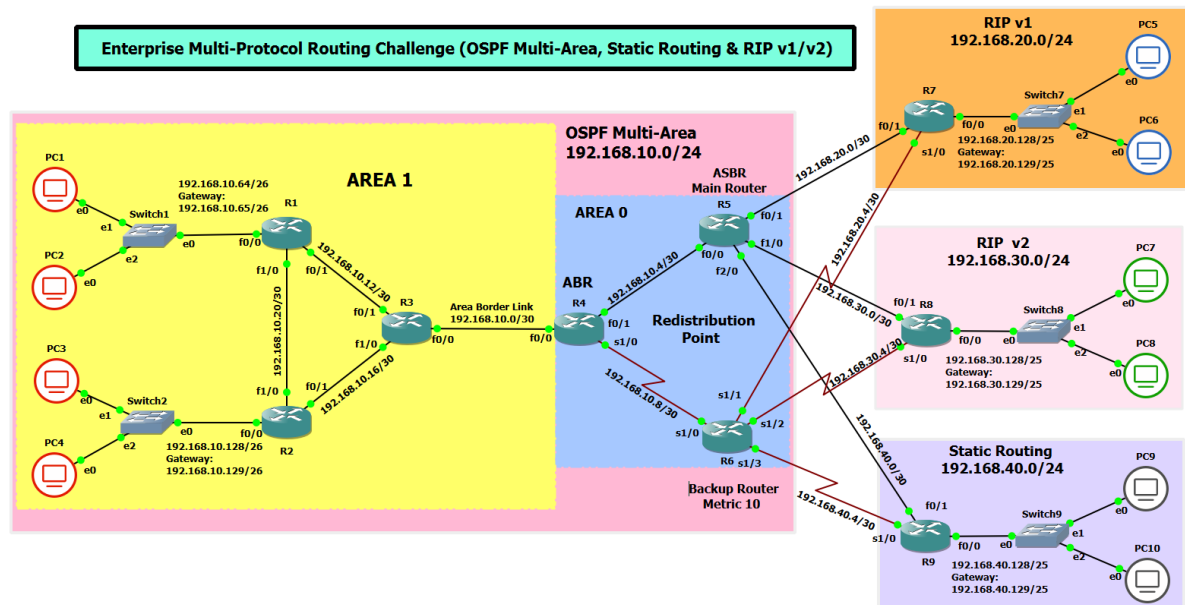
- **Definition:** RIP is a distance-vector routing protocol that uses hop count as its metric to find the best path.
- **Explanation:** RIP v1 is a classful protocol that does not support Variable Length Subnet Masking (VLSM). RIP v2 addresses this limitation by supporting classless routing, allowing subnet masks to be shared in routing updates, and using multicast (224.0.0.9) instead of broadcasts.

1.3. Route Redistribution

- **Definition:** The process of taking routing information from one routing protocol (e.g., RIP) and advertising it into another routing protocol (e.g., OSPF).
- **Explanation:** Since different protocols use different metrics (e.g., OSPF uses cost; RIP uses hop count), redistribution translates these metrics so that distinct routing domains can achieve end-to-end communication.

1.4. Floating Static Routes

- **Definition:** A backup static route configured with a higher Administrative Distance (AD) than the primary dynamic or static route.
- **Explanation:** The routing table only selects the path with the lowest AD. If the primary path goes down, the floating static route automatically populates the routing table to maintain network uptime.



2. Configuration & Verification Guide

2.1. IP Addressing Table (VLSM)

#	Network / Subnet	Mask	Usable Range	Broadcast	Used By
1	192.168.10.64/26	255.255.255.192	.65 – .126	.127	R1 LAN (fa0/0), gateway (.65), PC1 (.66), PC2 (.67)
2	192.168.10.128/26	255.255.255.192	.129 – .190	.191	R2 LAN (fa0/0), gateway (.129), PC3 (.130), PC4 (.131)
3	192.168.10.20/30	255.255.255.252	.21 – .22	.23	R1 fa1/0 (.21) – R2 fa1/0 (.22),
4	192.168.10.12/30	255.255.255.252	.13 – .14	.15	R1 fa0/1 (.13) – R3 fa0/1 (.14)
5	192.168.10.16/30	255.255.255.252	.17 – .18	.19	R2 fa0/1 (.17) – R3 fa1/0 (.18)
6	192.168.10.0/30	255.255.255.252	.1 – .2	.3	R3 fa0/0 (.1) – R4 fa0/0 (.2) (Area border link)

7	192.168.10.4/30	255.255.255.252	.5 – .6	.7	R4 fa0/1 (.5) – R5 fa0/0 (.6) (<i>Main, metric 1</i>)
8	192.168.10.8/30	255.255.255.252	.9 – .10	.11	R4 s1/0 (.9) – R6 s1/0 (.10) (<i>Backup, metric 10</i>)
9	192.168.20.0/30	255.255.255.252	.1 – .2	.3	R5 fa0/1 (.2) – R7 fa0/1 (.1)
10	192.168.20.4/30	255.255.255.252	.5 – .6	.7	R6 s1/1 (.6) – R7 s1/0 (.5)
11	192.168.30.0/30	255.255.255.252	.1 – .2	.3	R5 fa1/0 (.2) – R8 fa0/1 (.1)
12	192.168.30.4/30	255.255.255.252	.5 – .6	.7	R6 s1/2 (.6) – R8 s1/0 (.5)
13	192.168.40.0/30	255.255.255.252	.1 – .2	.3	R5 f2/0 (.2) – R9 fa0/1 (.1)
14	192.168.40.4/30	255.255.255.252	.5 – .6	.7	R6 s1/3 (.6) – R9 s1/0 (.5)
15	192.168.20.128/25	255.255.255.128	.129 – .254	.255	R7 LAN (fa0/0), gateway .129, PC5 (.130), PC6 (.131)
16	192.168.30.128/25	255.255.255.128	.129 – .254	.255	R8 LAN (fa0/0), gateway .129, PC7 (.130), PC8 (.131)
17	192.168.40.128/25	255.255.255.128	.129 – .254	.255	R9 LAN (fa0/0), gateway .129, PC9 (.130), PC10 (.131)

2.2. OSPF Neighbor Verification

To ensure that routers are communicating and exchanging topology data within the OSPF multi-area design:

- **Command:** show ip ospf neighbor
- **Explanation:** This verifies that neighbors have reached the FULL state, indicating successful database synchronization.

OSPF Neighbor Output

Router 3:

```
R3#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
2.2.2.2	1	FULL/DR	00:00:35	192.168.10.17	FastEthernet1/0
1.1.1.1	1	FULL/DR	00:00:36	192.168.10.13	FastEthernet0/1
4.4.4.4	1	FULL/BDR	00:00:35	192.168.10.2	FastEthernet0/0

Router 4:

```
R4#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
5.5.5.5	1	FULL/BDR	00:00:31	192.168.10.6	FastEthernet0/1
6.6.6.6	0	FULL/ -	00:00:35	192.168.10.10	Serial1/0
3.3.3.3	1	FULL/DR	00:00:30	192.168.10.1	FastEthernet0/0

(Note: Neighbor verification tables of R1, R2, R5 and R6 are uploaded in screenshots folder)

2.3. Routing Table & Redistribution Verification

To confirm that OSPF, RIP, and Static routes are correctly redistributing across the network:

- **Command:** show ip route
- **Explanation:** Check for codes like O IA (OSPF Inter-Area) and R (RIP) to ensure external routes are appearing across different domains.

Routing Table Snapshot:

Router 5:

 R5

```
R5#show ip route
```

```
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```
    192.168.30.0/24 is variably subnetted, 3 subnets, 2 masks
O E2   192.168.30.4/30 [110/10] via 192.168.10.5, 00:04:37, FastEthernet0/0
C       192.168.30.0/30 is directly connected, FastEthernet1/0
O E2   192.168.30.128/25 [110/10] via 192.168.10.5, 00:04:37, FastEthernet0/0
    192.168.10.0/24 is variably subnetted, 9 subnets, 3 masks
O IA    192.168.10.64/26 [110/32] via 192.168.10.5, 00:04:29, FastEthernet0/0
O IA    192.168.10.0/30 [110/20] via 192.168.10.5, 00:04:37, FastEthernet0/0
O E2    192.168.10.0/24 [110/10] via 192.168.10.5, 00:04:39, FastEthernet0/0
C       192.168.10.4/30 is directly connected, FastEthernet0/0
O       192.168.10.8/30 [110/74] via 192.168.10.5, 00:04:39, FastEthernet0/0
O IA    192.168.10.12/30 [110/30] via 192.168.10.5, 00:04:39, FastEthernet0/0
O IA    192.168.10.16/30 [110/21] via 192.168.10.5, 00:04:39, FastEthernet0/0
O IA    192.168.10.20/30 [110/22] via 192.168.10.5, 00:04:31, FastEthernet0/0
O IA    192.168.10.128/26 [110/31] via 192.168.10.5, 00:04:33, FastEthernet0/0
    192.168.40.0/24 is variably subnetted, 4 subnets, 3 masks
C       192.168.40.0/30 is directly connected, FastEthernet2/0
O E2    192.168.40.0/24 [110/10] via 192.168.10.5, 00:03:28, FastEthernet0/0
O E2    192.168.40.4/30 [110/10] via 192.168.10.5, 00:04:42, FastEthernet0/0
S       192.168.40.128/25 [1/0] via 192.168.40.1
    192.168.20.0/24 is variably subnetted, 4 subnets, 3 masks
O E2    192.168.20.4/30 [110/10] via 192.168.10.5, 00:04:43, FastEthernet0/0
C       192.168.20.0/30 is directly connected, FastEthernet0/1
O E2    192.168.20.0/24 [110/10] via 192.168.10.5, 00:03:30, FastEthernet0/0
O E2    192.168.20.128/25 [110/10] via 192.168.10.5, 00:04:45, FastEthernet0/0
```

Router 6:

R6

```
R6#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    192.168.30.0/24 is variably subnetted, 3 subnets, 2 masks
C       192.168.30.4/30 is directly connected, Serial1/2
O E2    192.168.30.0/30 [110/1] via 192.168.10.9, 00:02:53, Serial1/0
R       192.168.30.128/25 [120/1] via 192.168.30.5, 00:00:53, Serial1/2
    192.168.10.0/24 is variably subnetted, 9 subnets, 3 masks
O IA    192.168.10.64/26 [110/86] via 192.168.10.9, 00:02:53, Serial1/0
O IA    192.168.10.0/30 [110/74] via 192.168.10.9, 00:03:33, Serial1/0
R       192.168.10.0/24 [120/1] via 192.168.30.5, 00:00:55, Serial1/2
O       192.168.10.4/30 [110/74] via 192.168.10.9, 00:03:05, Serial1/0
C       192.168.10.8/30 is directly connected, Serial1/0
O IA    192.168.10.12/30 [110/84] via 192.168.10.9, 00:03:05, Serial1/0
O IA    192.168.10.16/30 [110/75] via 192.168.10.9, 00:03:05, Serial1/0
O IA    192.168.10.20/30 [110/76] via 192.168.10.9, 00:02:55, Serial1/0
O IA    192.168.10.128/26 [110/85] via 192.168.10.9, 00:02:57, Serial1/0
    192.168.40.0/24 is variably subnetted, 4 subnets, 3 masks
O E2    192.168.40.0/30 [110/1] via 192.168.10.9, 00:02:58, Serial1/0
R       192.168.40.0/24 [120/1] via 192.168.30.5, 00:00:02, Serial1/2
C       192.168.40.4/30 is directly connected, Serial1/3
S       192.168.40.128/25 [1/0] via 192.168.40.5
    192.168.20.0/24 is variably subnetted, 4 subnets, 3 masks
C       192.168.20.4/30 is directly connected, Serial1/1
O E2    192.168.20.0/30 [110/1] via 192.168.10.9, 00:03:01, Serial1/0
R       192.168.20.0/24 [120/1] via 192.168.30.5, 00:00:06, Serial1/2
S       192.168.20.128/25 [10/0] via 192.168.20.5
```

(Note: Other routers ip route verification is uploaded in screenshots folder)

2.4. Ping Test and Trace Route Results:

Ping Test Analysis: The initial timeouts shown in the sequences above are normal and happen because of the time it takes for ARP to resolve across multiple routing domains. Once resolved, the pings achieved a 100% success rate with stable response times across OSPF, RIP, and static boundaries.

Ping Test of PC1 to PC4, PC5, PC7, PC9:


```
PC1> ping 192.168.10.130
84 bytes from 192.168.10.130 icmp_seq=1 ttl=62 time=65.771 ms
84 bytes from 192.168.10.130 icmp_seq=2 ttl=62 time=64.766 ms
84 bytes from 192.168.10.130 icmp_seq=3 ttl=62 time=63.549 ms
84 bytes from 192.168.10.130 icmp_seq=4 ttl=62 time=63.786 ms
84 bytes from 192.168.10.130 icmp_seq=5 ttl=62 time=48.971 ms

PC1> ping 192.168.40.130
84 bytes from 192.168.40.130 icmp_seq=1 ttl=58 time=176.627 ms
84 bytes from 192.168.40.130 icmp_seq=2 ttl=58 time=178.172 ms
84 bytes from 192.168.40.130 icmp_seq=3 ttl=58 time=179.802 ms
84 bytes from 192.168.40.130 icmp_seq=4 ttl=58 time=161.638 ms
84 bytes from 192.168.40.130 icmp_seq=5 ttl=58 time=161.795 ms

PC1> ping 192.168.30.130
192.168.30.130 icmp_seq=1 timeout
192.168.30.130 icmp_seq=2 timeout
84 bytes from 192.168.30.130 icmp_seq=3 ttl=58 time=179.316 ms
84 bytes from 192.168.30.130 icmp_seq=4 ttl=58 time=176.998 ms
84 bytes from 192.168.30.130 icmp_seq=5 ttl=58 time=176.709 ms

PC1> ping 192.168.20.130
192.168.20.130 icmp_seq=1 timeout
192.168.20.130 icmp_seq=2 timeout
84 bytes from 192.168.20.130 icmp_seq=3 ttl=58 time=177.303 ms
84 bytes from 192.168.20.130 icmp_seq=4 ttl=58 time=145.477 ms
84 bytes from 192.168.20.130 icmp_seq=5 ttl=58 time=178.760 ms
```

Traceroute Path Mapping: The traceroute path illustrates the exact hops traffic takes through our enterprise topology:

- **Hop 1** (192.168.10.65): Local Default Gateway (Router 1 LAN).
- **Hop 2** (192.168.10.14): Moving into the OSPF Area 0 backbone.
- **Hop 3** (192.168.10.2): Area Border link crossing from Router 3 to Router 4.
- **Hop 4** (192.168.10.6 / .10): Core transit routers handling the distribution layer—Router 5 for the primary path and Router 6 for the secondary serial bypass path.
- **Hop 5:** Target edge routing interfaces serving the destination LAN segments (192.168.20.1, 192.168.30.5, or 192.168.40.1).

Tracer of PC1 to PC5, PC7, PC9:

```
PC1 - PuTTY
PC1> tracer 192.168.20.130
trace to 192.168.20.130, 8 hops max, press Ctrl+C to stop
 1  192.168.10.65    17.219 ms  17.024 ms  17.089 ms
 2  192.168.10.14   65.619 ms  65.973 ms  65.340 ms
 3  192.168.10.2    98.489 ms  98.026 ms  96.844 ms
 4  192.168.10.6    129.011 ms 130.390 ms 129.249 ms
 5  192.168.20.1    162.095 ms 161.834 ms 163.178 ms
 6  *192.168.20.130 178.678 ms (ICMP type:3, code:3, Desti

PC1> tracer 192.168.30.130
trace to 192.168.30.130, 8 hops max, press Ctrl+C to stop
 1  192.168.10.65    18.681 ms  17.425 ms  15.037 ms
 2  192.168.10.14   50.161 ms  66.322 ms  65.070 ms
 3  192.168.10.2    96.961 ms  98.301 ms  96.175 ms
 4  192.168.10.10   130.462 ms 130.030 ms 129.034 ms
 5  192.168.30.5    145.689 ms 161.782 ms 165.055 ms
 6  *192.168.30.130 177.238 ms (ICMP type:3, code:3, Desti

PC1> tracer 192.168.40.130
trace to 192.168.40.130, 8 hops max, press Ctrl+C to stop
 1  192.168.10.65    16.899 ms  16.026 ms  16.180 ms
 2  192.168.10.14   64.515 ms  65.248 ms  65.445 ms
 3  192.168.10.2    97.393 ms  97.576 ms  82.317 ms
 4  192.168.10.6    129.175 ms 131.336 ms 130.941 ms
 5  192.168.40.1    143.161 ms 163.503 ms 162.640 ms
 6  * * *
 7  *192.168.40.130 177.050 ms (ICMP type:3, code:3, Desti
```

3. Troubleshooting Report

3.1. Issue Description

- **Scenario:** Testing network redundancy and failover from Router 9 (R9) to Router 7 (R7).
- **Symptom:** When the primary link was intentionally disabled to test the backup route, traffic from R7 to R9 failed to reach its destination during a traceroute test.

3.2. Root Cause Analysis

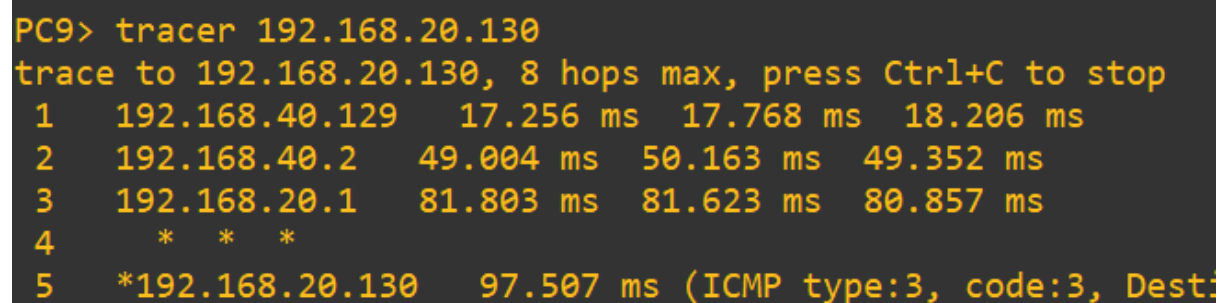
Upon reviewing the routing configuration during the simulated link failure, it was discovered that the explicit backup floating static route for the R9 destination subnet was missing from R7's configuration. Because the primary route dropped and no secondary static route existed for that specific backup path, traffic was discarded.

3.3. Resolution Steps

1. Identified the missing path to the destination network on R7.
2. Configured the backup static route explicitly on R7 pointing to the next-hop IP of the backup link.
3. Re-ran the traceroute command to confirm traffic successfully redirected through the backup path.

3.4. Verification:

Main Route Traceroute:



```
PC9> tracer 192.168.20.130
trace to 192.168.20.130, 8 hops max, press Ctrl+C to stop
 1  192.168.40.129    17.256 ms  17.768 ms  18.206 ms
 2  192.168.40.2     49.004 ms  50.163 ms  49.352 ms
 3  192.168.20.1     81.803 ms  81.623 ms  80.857 ms
 4      * * *
 5  *192.168.20.130  97.507 ms (ICMP type:3, code:3, Dest:192.168.20.130)
```

(Note; Main route is from R9 through R5 to R7)

Backup Route Traceroute:

```

PC9> tracer 192.168.20.130
trace to 192.168.20.130, 8 hops max, press Ctrl+C to stop
 1  192.168.40.129    26.564 ms  25.304 ms  9.207 ms
 2  192.168.40.6     50.866 ms  45.290 ms  43.533 ms
 3  192.168.20.5     56.825 ms  76.221 ms  81.208 ms
 4      *   *   *
 5  *192.168.20.130  89.901 ms (ICMP type:3, code:3, Destination
    unreachable)

PC9> ping 192.168.20.130
84 bytes from 192.168.20.130 icmp_seq=1 ttl=61 time=91.640 ms
84 bytes from 192.168.20.130 icmp_seq=2 ttl=61 time=101.689 ms
84 bytes from 192.168.20.130 icmp_seq=3 ttl=61 time=98.477 ms
84 bytes from 192.168.20.130 icmp_seq=4 ttl=61 time=95.824 ms
84 bytes from 192.168.20.130 icmp_seq=5 ttl=61 time=97.091 ms

```

(Note: After the main link was down, i.e: R9 through R5 to R7, Backup route from R9 through R6 to R7 was used)

4. Challenges Faced

4.1. Complex Multi-Protocol Redistribution on Core Routers (R5 & R6)

- **Challenge:** The primary obstacle during deployment was configuring seamless route redistribution between three distinct routing environments—OSPF, RIP, and Static Routing—simultaneously on the core transit routers, Router 5 and Router 6. Managing the translation of OSPF link-state metrics, RIP hop counts, and static paths on these main bypass routers required precise metric mapping to prevent routing loops or suboptimal routing paths.
- **Resolution:** Carefully mapped the redistribution metrics (assigning seed metrics for RIP and explicit costs/subnets tags for OSPF) on R5 and R6 to ensure that routes from all three domains were correctly injected and propagated across the entire network topology.

4.2. Static Routing Link Redundancy Failover (R7 to R9)

- **Challenge:** While testing the network's resilience, disabling the primary path between Router 7 and Router 9 caused traffic to drop entirely rather than gracefully failing over to the backup link.
- **Resolution:** Upon inspecting the routing tables during the link outage, it was determined that the floating static route configuration was missing the necessary backup

route entry on R7. Adding the explicit static route for the backup path successfully restored traffic flow, which was verified via a successful traceroute.

5. Conclusion

The implementation successfully demonstrates multi-protocol integration using Multi-Area OSPF, RIP, and Static Routing. Through deliberate route redistribution and the deployment of backup paths, the network achieved comprehensive end-to-end connectivity. The troubleshooting exercise highlighted the importance of verifying both primary and backup routing table entries to ensure seamless network resilience during a link failure.

